Project Proposal – 23CSE401 – Foundations of Artificial Intelligence

Project Title

Multi-Agent AI-Based Cybersecurity Threat Detection and Response System

Team Member Details

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Problem Statement

With the increasing number of cyber threats such as malware, phishing attacks, unauthorized access, and network intrusions, traditional security systems often struggle to analyze large amounts of security data and respond quickly. A single AI system handling all cybersecurity tasks can become complex and less efficient.

This project aims to develop a Multi-Agent AI-based cybersecurity system where multiple intelligent agents collaborate to detect, analyze, and respond to cyber threats. Each agent specializes in a specific security task such as intrusion detection, malware analysis, threat intelligence, risk assessment, and response planning.

Why Multi-Agent AI is Required for This System

Cybersecurity is a complex domain involving multiple independent tasks that require different types of analysis and decision-making. Using a single AI agent for all tasks makes the system difficult to maintain, less scalable, and harder to interpret.

A Multi-Agent AI approach is required because each agent can specialize in a specific cybersecurity function and collaborate with other agents to achieve a common goal. For example, a Network Monitoring Agent can identify suspicious traffic, a Malware Detection Agent can analyze malicious files, a Threat Intelligence Agent can provide knowledge about known attacks, and a Response Agent can select appropriate defensive actions.

The multi-agent approach provides the following advantages:

- Specialization: Each agent focuses on a specific cybersecurity task.
- Collaboration: Agents exchange information and improve decision-making.
- Scalability: New security agents can be added without redesigning the complete system.
- Faster Response: Multiple agents can analyze different aspects of a threat simultaneously.
- Explainability: Decisions can be traced through individual agent contributions.